Pragyan K C

San Antonio, TX 78249 | 336-653-0353 | pragyan.kc@utsa.edu
Google Scholar (50+ citations; h-index 3) | GitHub | Website

RESEARCH PROFILE

Research Summary. My research lies at the intersection of software engineering, requirements engineering, software evolution, and human-centered artificial intelligence. I develop AI-assisted methods using large language models, natural language processing, retrieval-augmented reasoning, and empirical software engineering to improve requirements quality, software traceability, testing, regulatory compliance, and trustworthy software evolution. My long-term agenda focuses on intelligent human-in-the-loop developer-support systems that preserve stakeholder intent and enable reliable, accountable, AI-assisted software engineering.

Research Interests. Software Engineering; Requirements Engineering; AI for Software Engineering; Large Language Models; Software Evolution; Software Traceability; Software Testing; Human-Centered AI; Privacy and Regulatory Compliance

EDUCATION

Ph.D., Computer Science - The University of Texas at San Antonio **Expected Spring 2027**
- Dissertation area: Requirements Engineering, AI-assisted Software Engineering, and LLMs for Software Artifacts; Advisor: Dr. Mitra Bokaei Hosseini

M.Sc., Computer Science - The University of Texas at San Antonio **Spring 2026**

B.S., Computer Science (Data Science Track) - St. Mary's University **Spring 2022**

ACADEMIC APPOINTMENTS AND RESEARCH EXPERIENCE

Visiting Researcher - Carnegie Mellon University **Summer 2026**
Supervisor: Dr. Travis Breaux
- Developed a goal-conditioned, LLM-based approach that extracts stakeholder goals from real-world issues (SWE-bench Verified, SWE-bench-Live) to guide generation of patches and test cases.
- Investigated whether goal conditioning broadens the behavioral coverage of generated patches and tests while preserving traceability from goals to code and validation artifacts.

Graduate Teaching Assistant II - UT San Antonio **Fall 2025 - Present**
- Served as Instructor of Record for CS 2113 (Fall 2025) and CS 3443 (Spring 2026) under this appointment; currently supports undergraduate and graduate data science courses offered in both in-person and online sections.

Research Assistant I - Data Driven Software Engineering Lab, UT San Antonio **Fall 2022 - Summer 2025**
Supervisor: Dr. Mitra Bokaei Hosseini
- Developed AI-assisted software engineering methods using large language models, natural language processing, retrieval-augmented reasoning, and empirical software engineering techniques.
- Conducted empirical studies of feature requests, issue trackers, developer workflows, and community-driven software evolution in open-source ecosystems.
- Developed and curated datasets involving privacy policies, user scenarios, feature requests, issue trackers, and community-forum discussions; conducted interviews, surveys, and mixed-methods analyses.
- Managed research infrastructure and led interdisciplinary reading groups and technical discussions on software engineering and AI-assisted development.

Graduate Research Intern - International Computer Science Institute **Summer 2024**
Supervisor: Dr. Primal Wijesekera
- Investigated privacy risks in telehealth applications and designed an IRB-approved study involving privacy-aware software systems and stakeholder analysis.

Undergraduate Research Assistant - St. Mary's University **Fall 2020 - Summer 2021**
Supervisor: Dr. Mitra Bokaei Hosseini
- Conducted NLP-based research on privacy-policy analysis and third-party entity identification; assisted with data collection, annotation, and machine-learning experiments.

PEER-REVIEWED PUBLICATIONS

Conference Papers

1. **Pragyan K C**, Mitra Bokaei Hosseini. “From Community Forums to Issue Trackers: A Moodle Case Study.” IEEE International Requirements Engineering Conference (RE), 2026.
2. **Pragyan K C**, Rambod Ghandiparsi, Thomas Herron, John Heaps, Mitra Bokaei Hosseini. “Demystifying Feature Requests: Leveraging LLMs to Refine Feature Requests in Open-Source Software.” IEEE RE, 2025.

3. Gabriel Morales*, **Pragyan K C***, Sadia Jahan, Mitra Bokaei Hosseini, Rocky Slavin. “A Large-Language-Model Approach to Code and Privacy Policy Alignment.” IEEE SANER, 2024. *Co-first authors.

Workshop Papers

4. Rambod Ghandiparsi, Oluwaseyi Lawal, **Pragyan K C**, Mitra Bokaei Hosseini. “Regression Bugs in Post-Release GitHub Issue Reports: A Pilot Study.” CrowdRE at IEEE RE, 2026.
5. **Pragyan K C**, Rambod Ghandiparsi, Thomas Herron, John Heaps, Mitra Bokaei Hosseini. “Towards Better Requirements from the Crowd: Developer Engagement with Feature Requests in Open Source Software.” CrowdRE at IEEE RE, 2025.
6. **Pragyan K C**, Rambod Ghandiparsi, Rocky Slavin, Sepideh Ghanavati, Travis Breaux, Mitra Bokaei Hosseini. “Toward Regulatory Compliance: A Few-Shot Learning Approach to Extract Processing Activities.” IEEE REW, 2024.
7. Mitra Bokaei Hosseini, **Pragyan K C**, Irwin Reyes, Serge Eagleman. “Identifying and Classifying Third-party Entities in Natural Language Privacy Policies.” Privacy in NLP at EMNLP, 2020.

Doctoral Symposium / Extended Abstract

8. **Pragyan K C**. “Requirements Copilot: Ambiguity Management in Feature Requests.” IEEE RE Doctoral Symposium, 2024.

FELLOWSHIP AND PROPOSAL ACTIVITY

Google PhD Fellowship - Institutional Nominee 2026, The University of Texas at San Antonio

• Research proposal selected through UT San Antonio's limited internal review process for institutional submission; the university may submit no more than four proposals.

TEACHING EXPERIENCE

Graduate Teaching Assistant II - Data Science, UT San Antonio **Fall 2026**

- Introduction to Data Science: DS-4003-001, DS-4003-ON1, DS-5003-001, DS-5003-ON1.
- Programming for Data Science: DS-4013-002, DS-4013-ON2, DS-5013-002, and DS-5013-ON3.
- Data Organization & Visualization: DS-5023-001, DS-5023-ON1, DS-4023-ON1, and DS-4023-001.

Instructor of Record - CS 3443: Application Programming, UT San Antonio **Spring 2026**

48 students

- Independently taught object-oriented programming and software development practices including UML modeling, MVC architecture, Git/GitHub workflows, software testing, multithreading, database integration, and JavaFX application development.
- Designed project-based learning experiences emphasizing modular design, code quality, testing, documentation, debugging, and iterative software development.
- Teaching evaluations: 4.68/5 overall; 4.80/5 for availability outside class; 4.75/5 for communication effectiveness.

Instructor of Record - CS 2113: Object-Oriented Programming, UT San Antonio **Fall 2025**

38 students

- Independently taught encapsulation, inheritance, polymorphism, abstraction, exception handling, class design, interfaces, recursion, and file I/O in Java, while introducing modular design, debugging, code organization, and testing.
- Teaching evaluations: 4.83/5 for course-objective clarity; 4.75/5 for instructional preparedness.

COURSES PREPARED TO TEACH

Software Engineering; Requirements Engineering; Software Architecture and Design; Software Testing and Quality Assurance; Object-Oriented Programming; Application Programming / Software Development; Artificial Intelligence (AI); AI for Software Engineering; Software Evolution and Maintenance; Programming for Data Science; Data Organization and Visualization

RESEARCH MENTORING

- Zihao Liu (Graduate Student) - Mentored development of a recursive parser and HTML-based annotation framework for user scenarios supporting event-argument detection, use-case extraction, and regulatory-compliance analysis.
- Maria Jose Velazquez (Graduate Student) - Mentored design and implementation of an LLM-assisted chatbot for detecting ambiguity and incompleteness in feature requests, extending work published at IEEE RE 2025.

RESEARCH PRESENTATIONS

- “From Community Forums to Issue Trackers: A Moodle Case Study.” IEEE RE 2026, Montreal, Canada.
- “From Community Forums to Issue Trackers: A Moodle Case Study.” Brown Bag Seminar, Department of Information Systems and Cyber Security, UT San Antonio, Spring 2026.
- “Demystifying Feature Requests: Leveraging LLMs to Refine Feature Requests in Open-Source Software.” IEEE RE 2025, Valencia, Spain.

- "Towards Better Requirements from the Crowd: Developer Engagement with Feature Requests in Open Source Software." CrowdRE at IEEE RE 2025, Valencia, Spain.
- "Toward Regulatory Compliance: A Few-Shot Learning Approach to Extract Processing Activities." ESPRE at IEEE RE 2024, Reykjavik, Iceland.
- "Requirements Copilot: Ambiguity Management in Feature Requests." IEEE RE Doctoral Symposium 2024, Reykjavik, Iceland.
- "A Large-Language-Model Approach to Code and Privacy Policy Alignment." IEEE SANER 2024, Rovaniemi, Finland.

PROFESSIONAL SERVICE

- Program Committee Member, RE@Next, IEEE Requirements Engineering, 2026.
- Program Committee Member, CrowdRE Workshop, IEEE Requirements Engineering, 2026.
- Program Committee Member, ESPRE Workshop, IEEE Requirements Engineering, 2026.
- Student Volunteer, IEEE Requirements Engineering, 2026.
- Program Committee Member, ESPRE Workshop, IEEE Requirements Engineering, 2025.
- Program Committee Member, AIRE Workshop, IEEE Requirements Engineering, 2025.
- Judge, Poster and Oral Presentations, Research Showcase, St. Mary's University, 2026.
- Judge, Semi-Annual Computer Science Symposium, St. Mary's University, Spring 2025.
- Student Volunteer, IEEE Requirements Engineering, 2025.

AWARDS AND HONORS

- UTSA Graduate Student Professional Development Award ($750), IEEE RE 2025.
- UTSA Graduate Student Professional Development Award ($750), IEEE SANER 2024.
- 2nd Place Poster, "Toward Regulatory Compliance: A Few-Shot Learning Approach to Extract Processing Activities," Los Datos Conference, Fall 2024.
- 3rd Place Poster, "Feature Request Apprentice - Managing Software Changes from Feature Requests," Los Datos Conference, Fall 2024.
- Presidential Award, St. Mary's University, 2022 - one of 14 graduating seniors selected for service, leadership, and academic excellence.
- Outstanding Senior, Computer Science, St. Mary's University, 2022.
- Best Senior Project, "Blood Bank Nepal," Semi-Annual Computer Science Symposium, St. Mary's University, 2022.
- 2nd Place Poster, "Automated Third-Party Name Elicitation from Natural Language Processing," Annual Research Showcase, 2021.
- St. Mary's University Summer Undergraduate Research Fellowship (SURF) Award ($2,000), 2020.

LEADERSHIP AND UNIVERSITY SERVICE

- Graduate School Ph.D. Advisory Committee Member, UT San Antonio, Spring 2026.
- HSC Student Career Planning Working Group Member, UT San Antonio, Fall 2025.
- New Computing College Task Force Member, UT San Antonio, Spring 2024.
- President, Computer Science Graduate Student Association, UT San Antonio, Fall 2023-Spring 2024.
- President, Nepalese Student Association, UT San Antonio, Fall 2025-Spring 2026.

TECHNICAL AND RESEARCH EXPERTISE

Programming and tools: Python, Java, JavaFX, Git/GitHub, Linux. **Methods:** Large language models, natural language processing, retrieval-augmented reasoning, software repository mining, qualitative and mixed-methods analysis, interviews and surveys, human-in-the-loop evaluation.